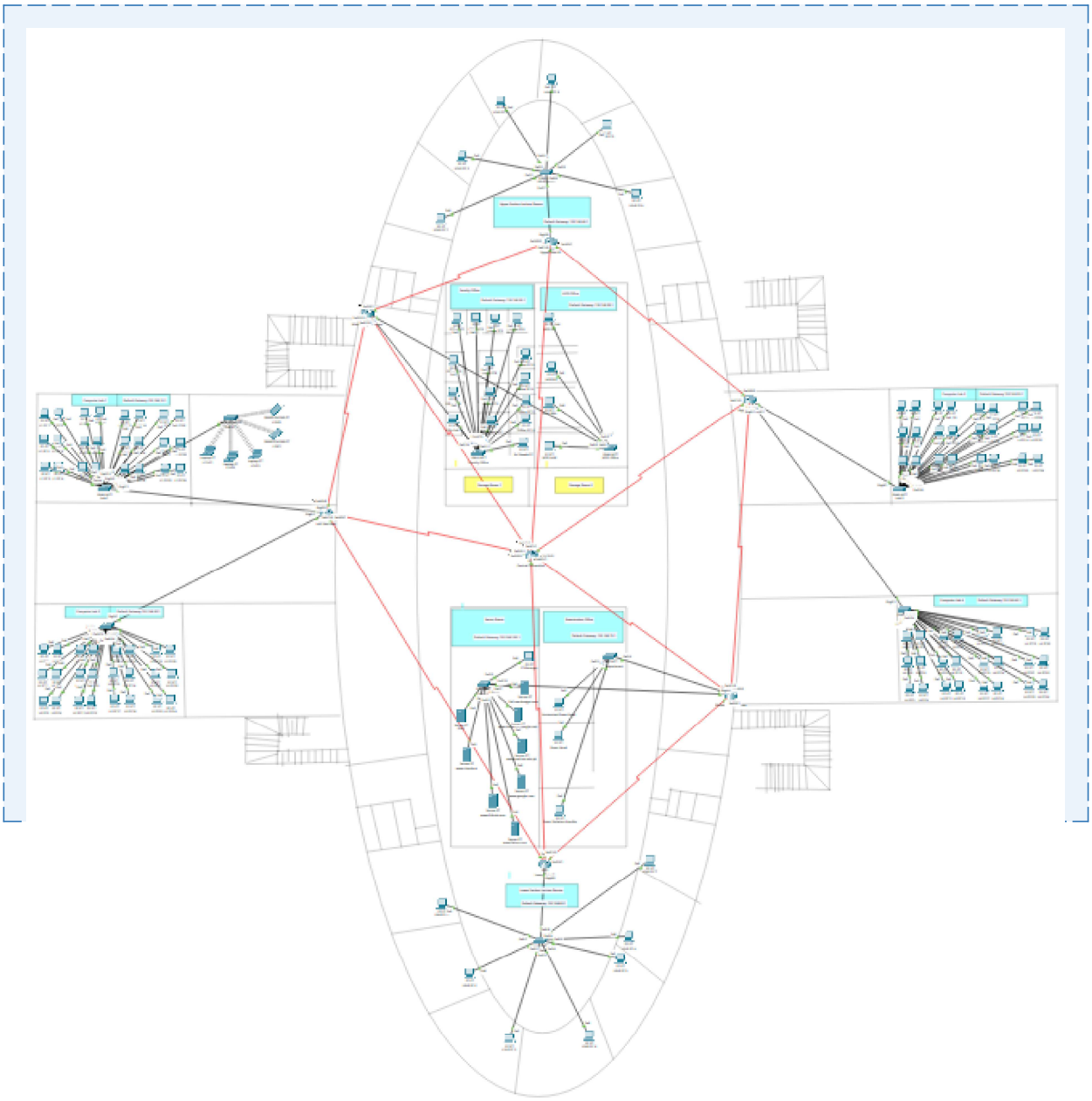
1. Executive Summary

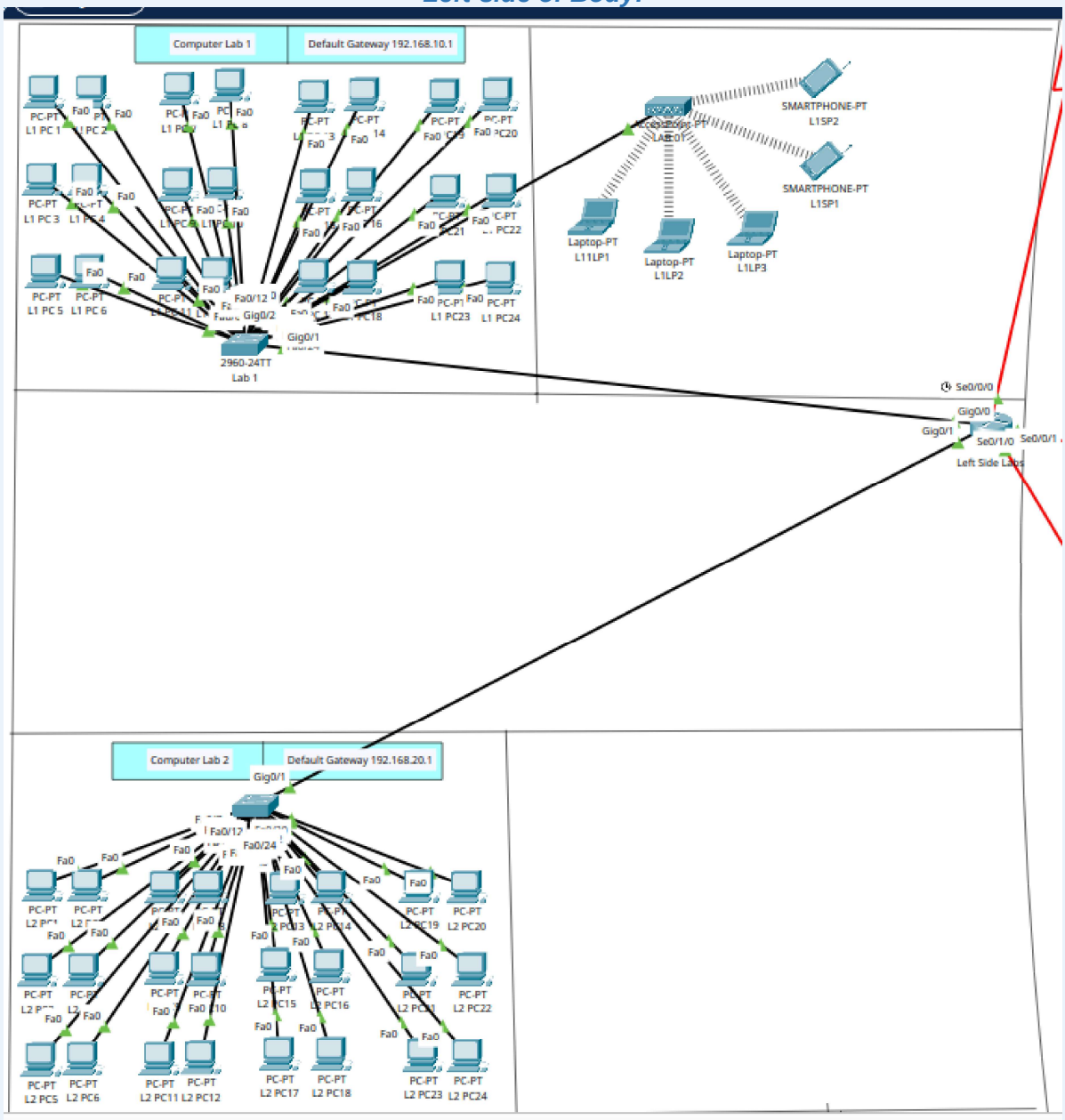
This report documents the design, configuration, and implementation of a fully functional university campus network using Cisco Packet Tracer. The network spans an entire university building and interconnects multiple zones including computer labs, lecture rooms, a faculty office, a HOD (Head of Department) office, an exam office, a server room, and a dedicated network infrastructure backbone.

The project demonstrates practical application of core networking concepts including static IP addressing, inter-zone routing, fault-tolerant ring topology, DNS resolution, HTTP web hosting, email services, Access Control Lists (ACLs), and wireless connectivity.

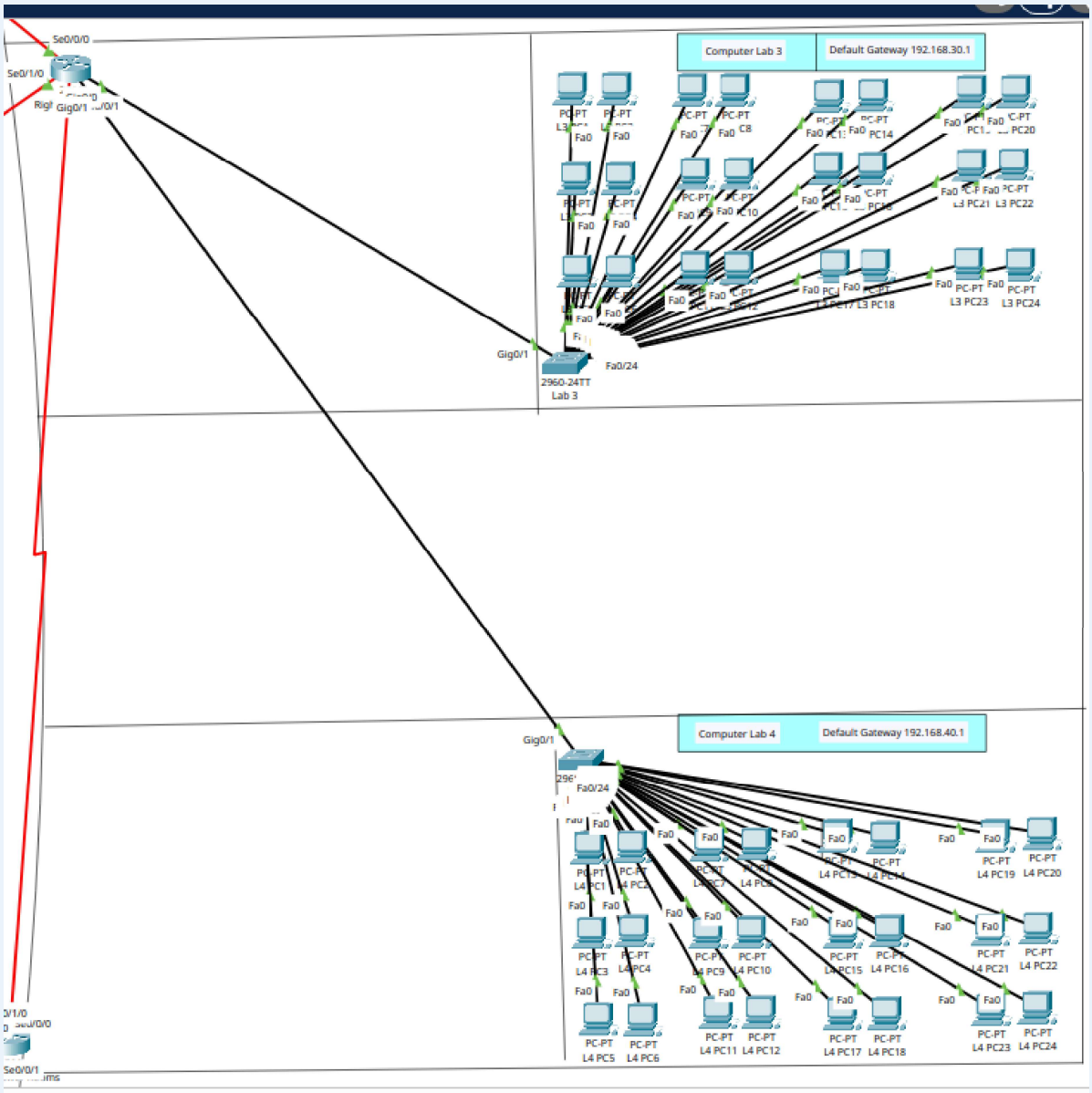


Full-Body Diagram

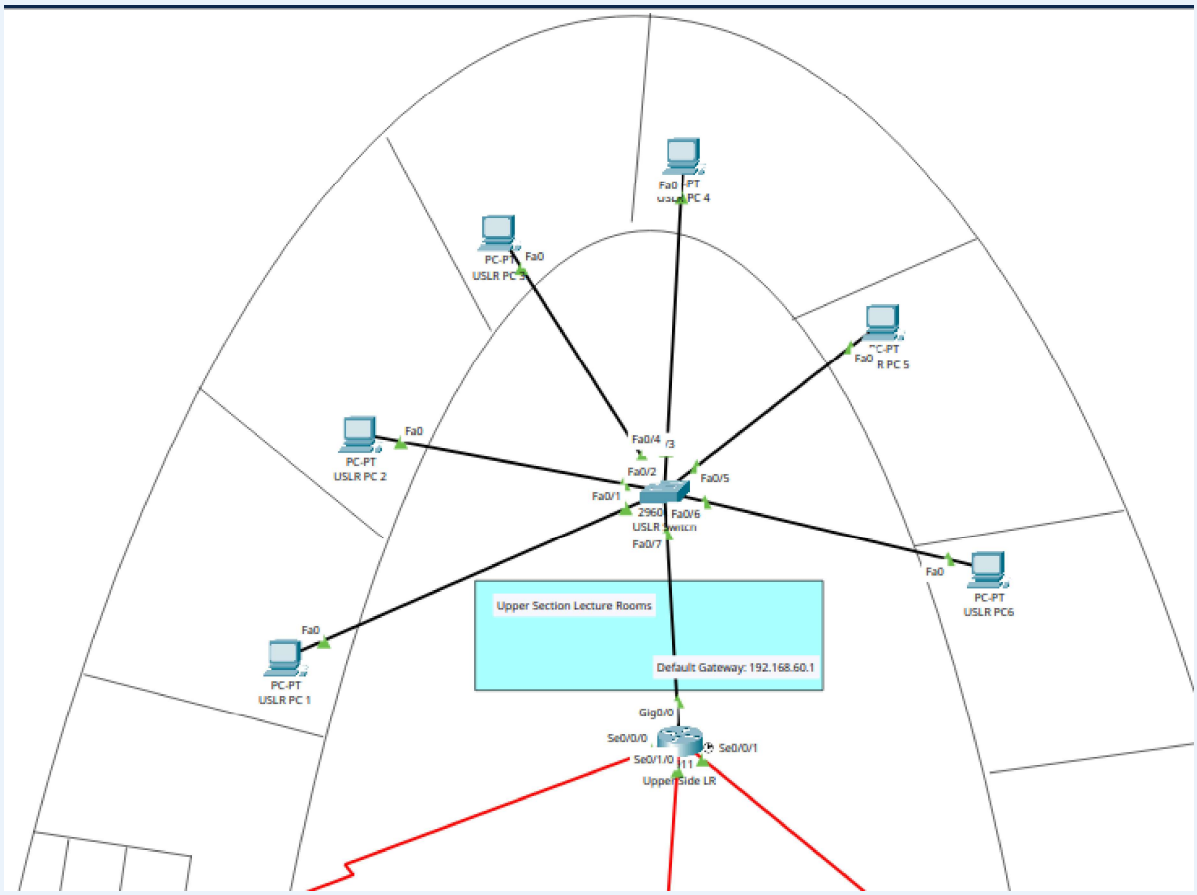
Left-side of Body:



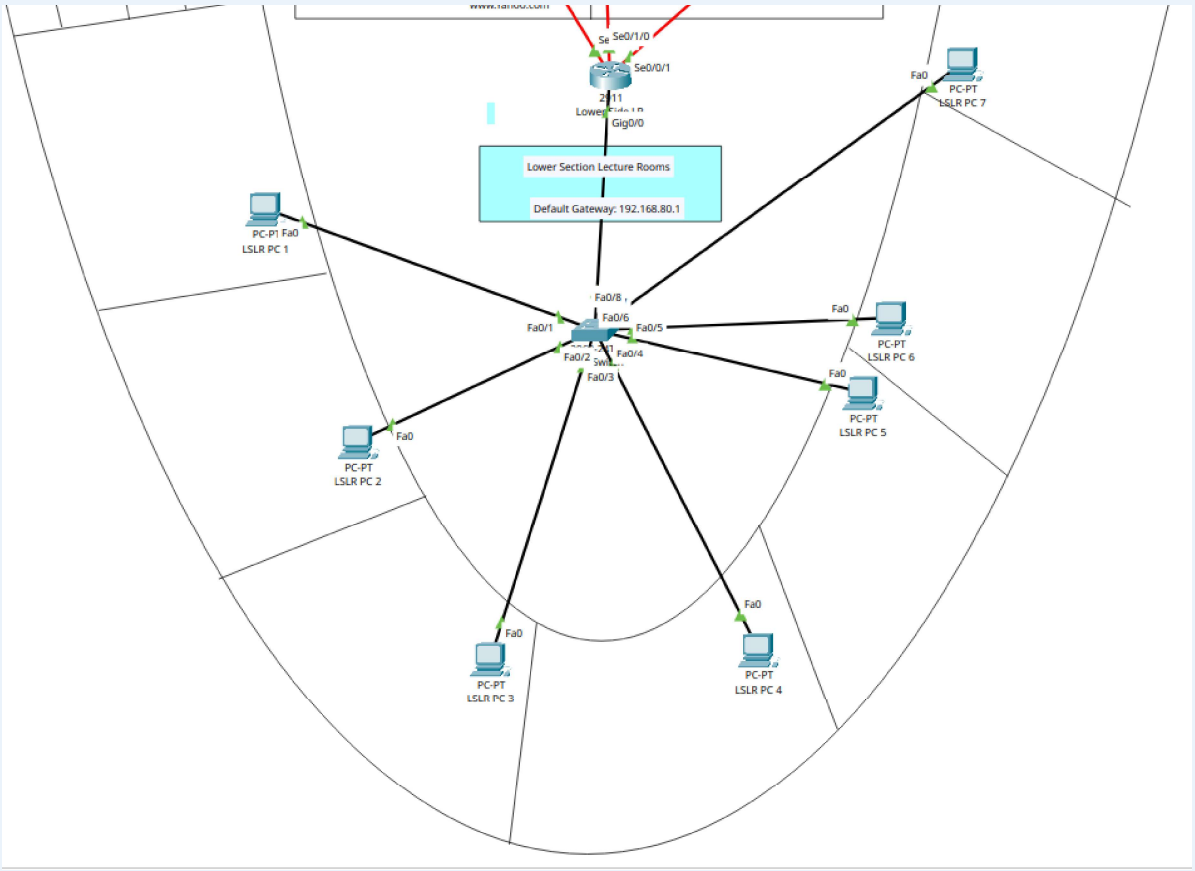
Right Side of Body:



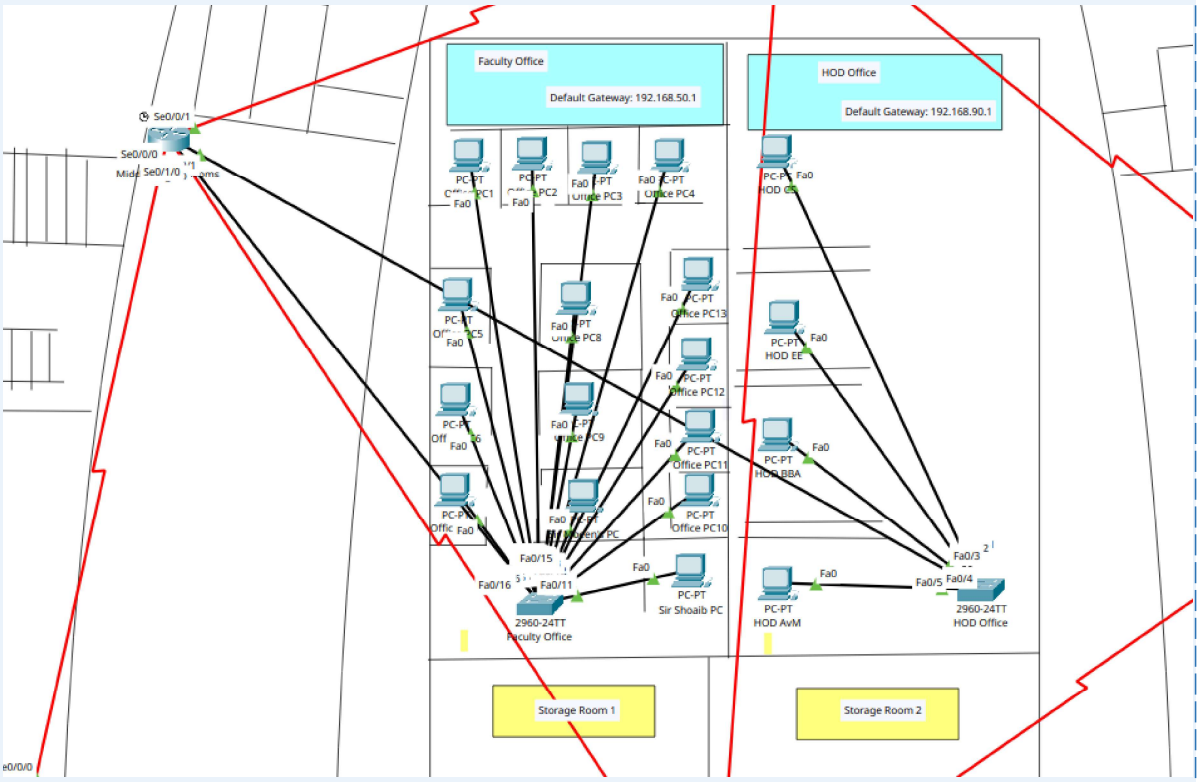
Top Side of Body:



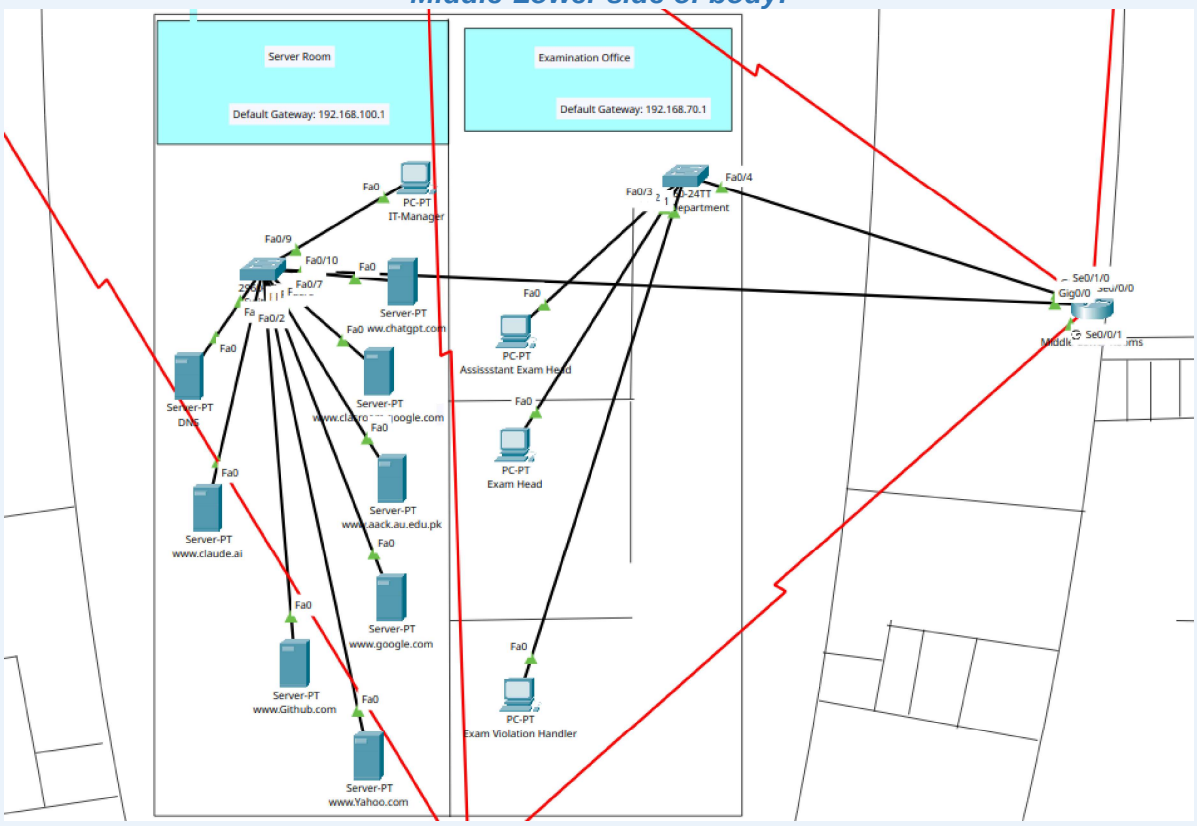
Bottom Side of body:



Middle-Upper Side of Body:



Middle-Lower side of body:



2. Project Objectives

The primary goals of this project were:

- Design a scalable, hierarchical campus network for a multi-zone university building
- Implement static IP addressing with clear subnet isolation per zone
- Deploy DNS and HTTP servers to simulate real-world web hosting
- Establish secure inter-zone communication using Access Control Lists (ACLs)
- Achieve fault-tolerant routing using a ring topology with floating static routes
- Configure email services for internal university communication
- Add wireless access points for mobile device connectivity

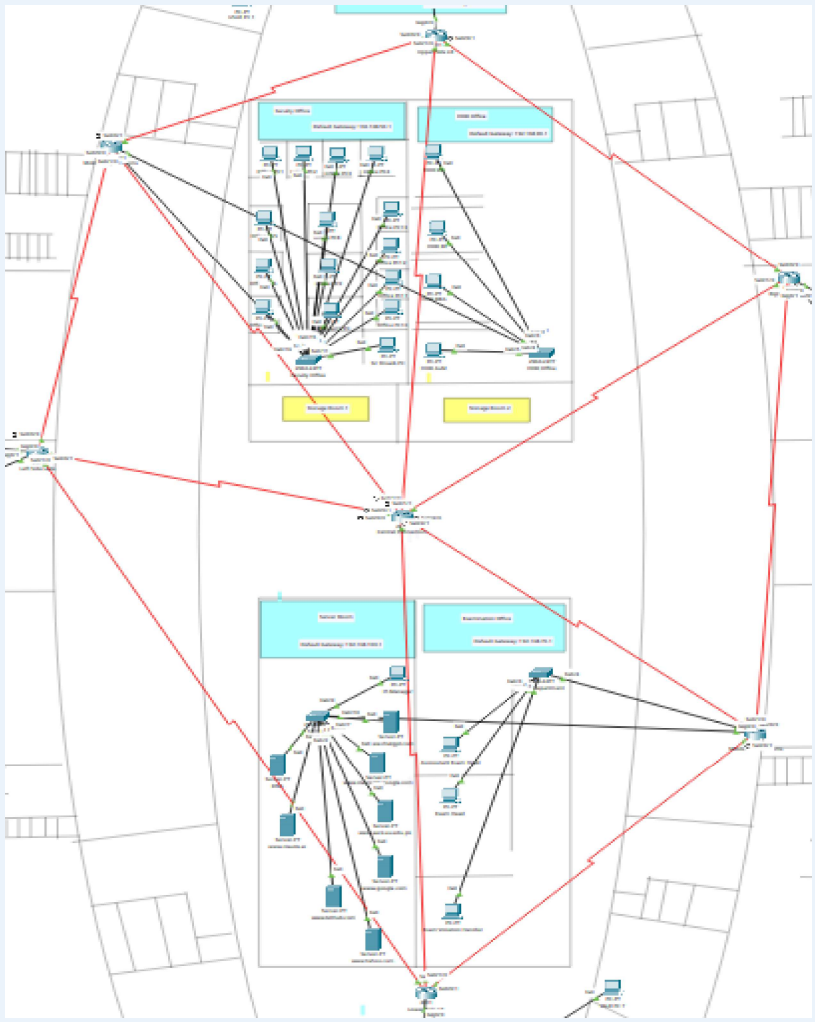
3. Network Topology Overview

The network follows a hybrid Star-Ring topology. A central router forms the backbone, while six zone routers are interconnected in a ring for redundancy. Each zone router manages its local switches and end devices.

3.1 Topology Design

The topology consists of two layers:

- Core Layer: Central Router (2911) connected via serial links to all 6 zone routers
- Distribution Layer: Six zone routers (2911 + HWIC-2T) connected in a ring using serial DCE cables
- Access Layer: Switches (Cisco 2960) connecting end devices within each zone



3.2 Zone Router Assignments

Router	Zone Covered	LAN Subnets	Serial Ring Links
Left Side	Lab 1 + Lab 2	192.168.10.0 / 192.168.20.0	↔ Mid-Upper, ↔ Bottom
Right Side	Lab 3 + Lab 4	192.168.30.0 / 192.168.40.0	↔ Top, ↔ Mid-Lower
Mid-Upper	Faculty Office + HOD	192.168.50.0 / 192.168.90.0	↔ Left, ↔ Top
Upper Side	Top Classrooms	192.168.60.0	↔ Mid-Upper, ↔ Right
Mid-Lower	Server Room + Exam Office	192.168.100.0 / 192.168.70.0	↔ Right, ↔ Bottom
Lower Side	Bottom Classrooms	192.168.80.0	↔ Mid-Lower, ↔ Left

4. IP Addressing Scheme

All devices use static IP addressing for reliability and ease of debugging. The network uses the 192.168.X.0/24 address space for LAN segments and 10.0.X.0/30 and 10.1.X.0/30 for router-to-router WAN links.

4.1 LAN Subnet Table

Zone	Subnet	Gateway	Subnet Mask
Lab 1 (Left)	192.168.10.0/24	192.168.10.1	255.255.255.0
Lab 2 (Left)	192.168.20.0/24	192.168.20.1	255.255.255.0
Lab 3 (Right)	192.168.30.0/24	192.168.30.1	255.255.255.0
Lab 4 (Right)	192.168.40.0/24	192.168.40.1	255.255.255.0
Faculty Office	192.168.50.0/24	192.168.50.1	255.255.255.0
Top Classrooms	192.168.60.0/24	192.168.60.1	255.255.255.0
Exam Office	192.168.70.0/24	192.168.70.1	255.255.255.0
Bottom Classrooms	192.168.80.0/24	192.168.80.1	255.255.255.0
HOD Office	192.168.90.0/24	192.168.90.1	255.255.255.0
Server Room	192.168.100.0/24	192.168.100.1	255.255.255.0

4.2 WAN / Router-to-Router Links

Point-to-point /30 subnets are used between routers, giving exactly 2 usable IP addresses per link — perfect for serial connections.

Link	Subnet	Router A IP	Router B IP
Central ↔ Left	10.1.1.0/30	10.1.1.1	10.1.1.2
Central ↔ Mid-Upper	10.1.2.0/30	10.1.2.1	10.1.2.2
Central ↔ Upper	10.1.3.0/30	10.1.3.1	10.1.3.2
Central ↔ Right	10.1.4.0/30	10.1.4.1	10.1.4.2
Central ↔ Mid-Lower	10.1.5.0/30	10.1.5.1	10.1.5.2
Central ↔ Lower	10.1.6.0/30	10.1.6.1	10.1.6.2

Ring: Left ↔ Mid-Upper	10.0.1.0/30	10.0.1.1	10.0.1.2
Ring: Mid-Upper ↔ Top	10.0.2.0/30	10.0.2.1	10.0.2.2
Ring: Top ↔ Right	10.0.3.0/30	10.0.3.1	10.0.3.2
Ring: Right ↔ Mid-Lower	10.0.4.0/30	10.0.4.1	10.0.4.2
Ring: Mid-Lower ↔ Bottom	10.0.5.0/30	10.0.5.1	10.0.5.2
Ring: Bottom ↔ Left	10.0.6.0/30	10.0.6.1	10.0.6.2

5. Routing Configuration

5.1 Routing Strategy

Static routing is used throughout the network. This approach was chosen over dynamic routing protocols (such as OSPF or RIP) for the following reasons:

- Predictable and deterministic packet paths
- Easier to debug in a lab environment
- No routing protocol overhead on the network
- Full control over primary and backup paths

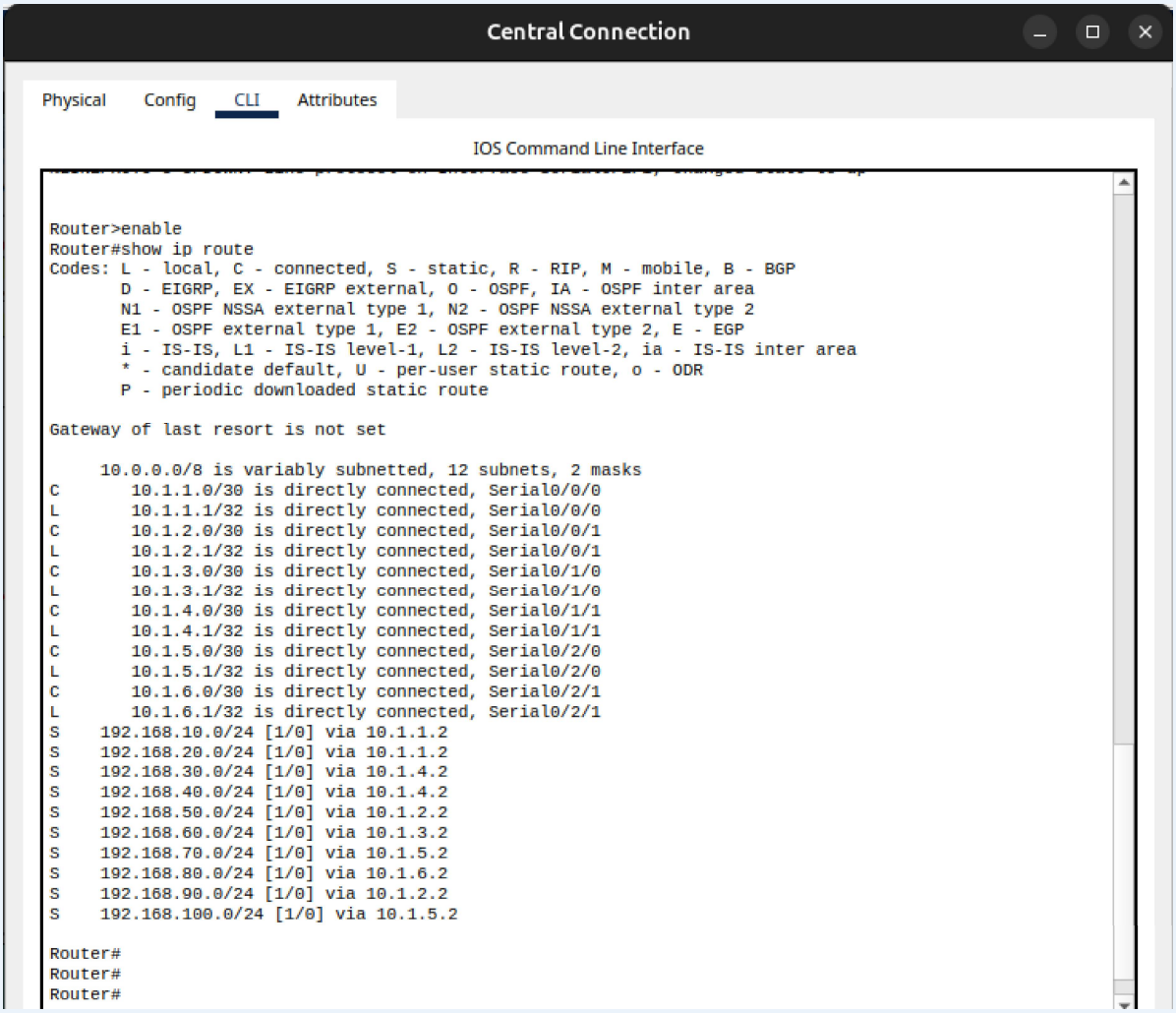
5.2 Floating Static Routes — Fault Tolerance

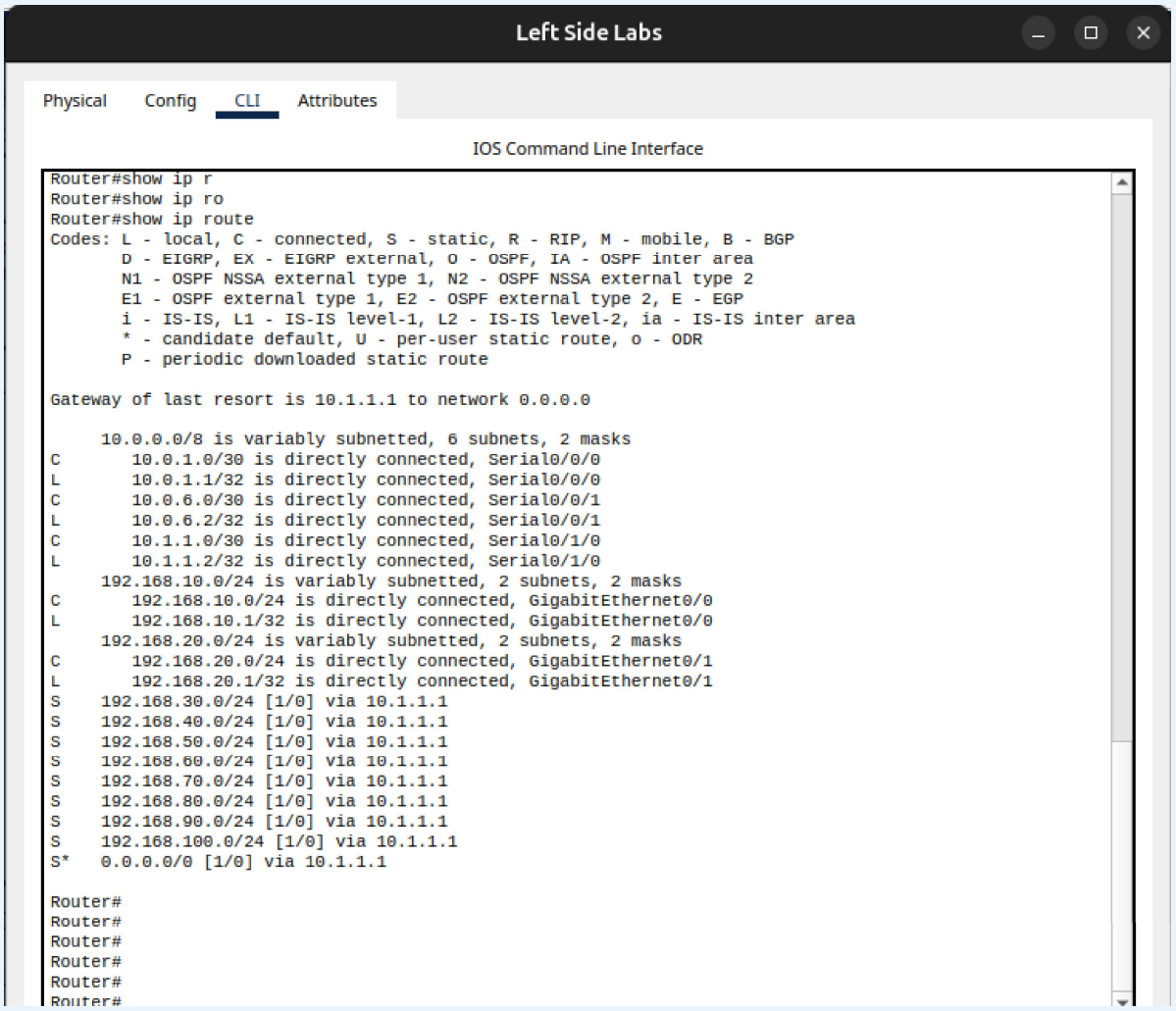
Each zone router has three levels of routing priority using Administrative Distance (AD) values:

AD Value	Path Type	Used When	Example
1	Primary — via Central	Central router is UP	All traffic goes through backbone
10	Backup 1 — via Ring Neighbor A	Central is DOWN	Traffic hops to nearest ring neighbor
20	Backup 2 — via Ring Neighbor B	Central + Neighbor A DOWN	Traffic takes opposite ring direction

📌 **NOTE:** Lower AD value = Higher Priority. The router always picks the path with the lowest AD. Floating routes only activate when a higher-priority path goes down.

The routing Table on Central Connection router


Central Connection
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C 10.1.1.0/30 is directly connected, Serial0/0/0
L 10.1.1.1/32 is directly connected, Serial0/0/0
C 10.1.2.0/30 is directly connected, Serial0/0/1
L 10.1.2.1/32 is directly connected, Serial0/0/1
C 10.1.3.0/30 is directly connected, Serial0/1/0
L 10.1.3.1/32 is directly connected, Serial0/1/0
C 10.1.4.0/30 is directly connected, Serial0/1/1
L 10.1.4.1/32 is directly connected, Serial0/1/1
C 10.1.5.0/30 is directly connected, Serial0/2/0
L 10.1.5.1/32 is directly connected, Serial0/2/0
C 10.1.6.0/30 is directly connected, Serial0/2/1
L 10.1.6.1/32 is directly connected, Serial0/2/1
S 192.168.10.0/24 [1/0] via 10.1.1.2
S 192.168.20.0/24 [1/0] via 10.1.1.2
S 192.168.30.0/24 [1/0] via 10.1.4.2
S 192.168.40.0/24 [1/0] via 10.1.4.2
S 192.168.50.0/24 [1/0] via 10.1.2.2
S 192.168.60.0/24 [1/0] via 10.1.3.2
S 192.168.70.0/24 [1/0] via 10.1.5.2
S 192.168.80.0/24 [1/0] via 10.1.6.2
S 192.168.90.0/24 [1/0] via 10.1.2.2
S 192.168.100.0/24 [1/0] via 10.1.5.2
Router#
Router#
Router#



The Left Side Labs Routers Direct and Backup paths

6. Server Room Configuration

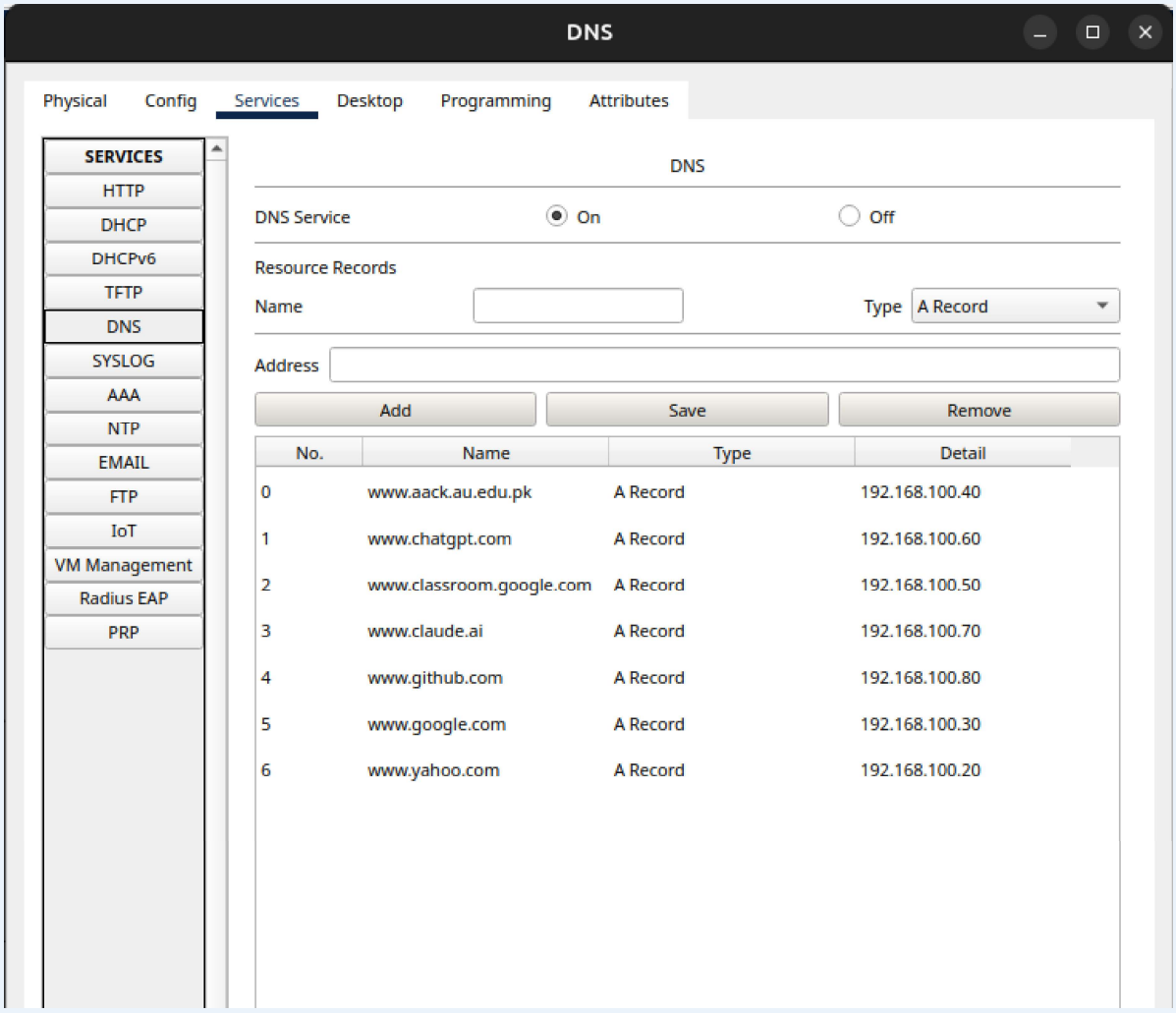
The server room is the core services hub of the campus network. It is connected to the Mid-Lower zone router and hosts all DNS and HTTP services. Any device on the network can access the server room without restrictions.

6.1 Server Inventory

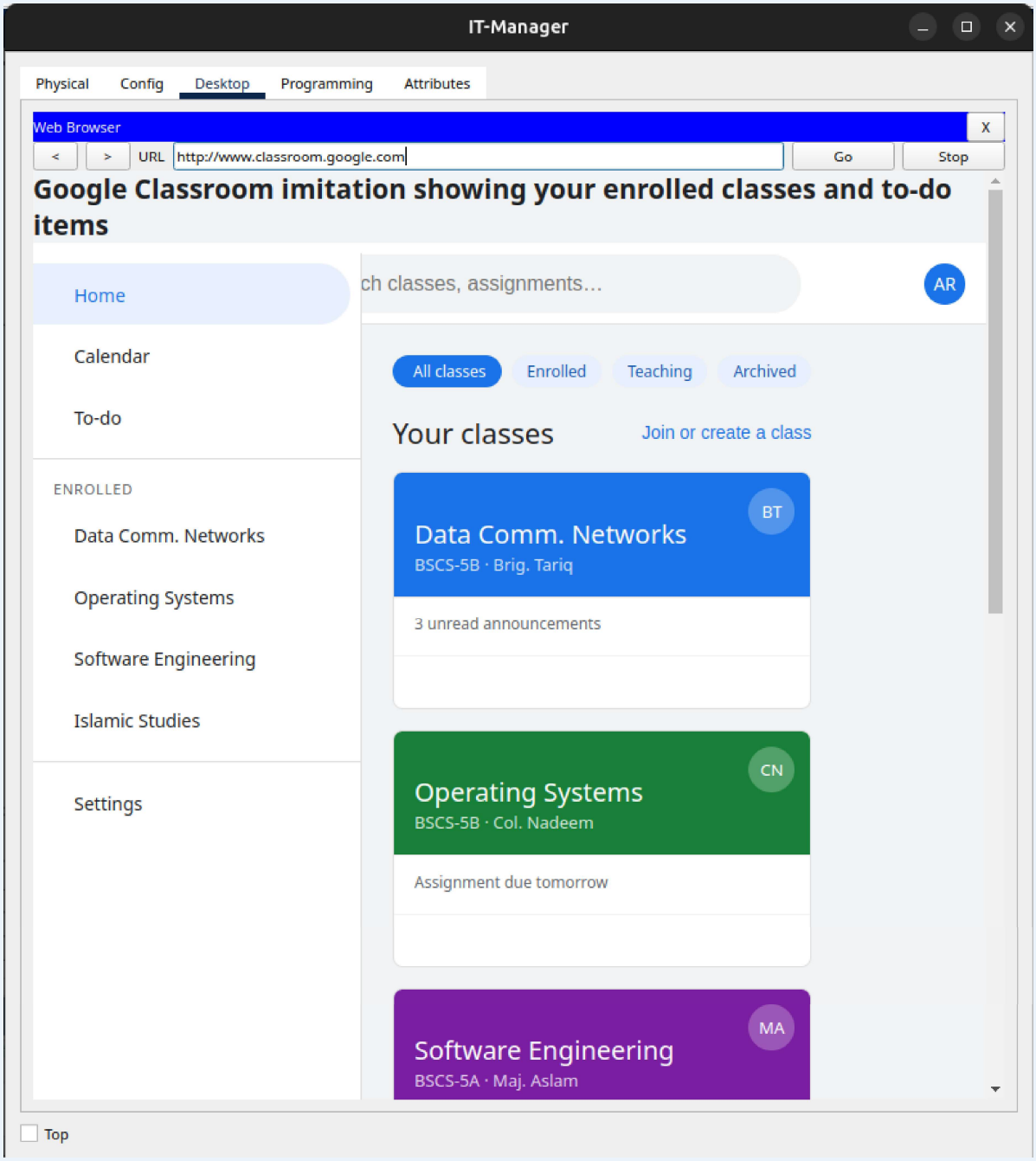
Device	Domain / Role	IP Address	Service
DNS Server	DNS Resolution	192.168.100.10	DNS
Server-PT	www.chatgpt.com	192.168.100.60	HTTP
Server-PT	www.claude.ai	192.168.100.70	HTTP
Server-PT	www.yahoo.com	192.168.100.20	HTTP
Server-PT	google.com	192.168.100.30	HTTP
Server-PT	www.classroom.google.com	192.168.100.50	HTTP
Server-PT	www.aack.au.edu.pk	192.168.100.40	HTTP
Server-PT	Email Server	192.168.100.10	SMTP/POP3
PC-PT	IT Manager	192.168.100.5	Management

6.2 DNS Configuration

The DNS server holds A-records mapping each domain name to its corresponding web server IP address. Every end device on the network is configured with the DNS server IP (192.168.100.2) to enable domain name resolution.



DNS Server Configuration



IT-Manager in Server Room fetches the Domain to check the workings.

7. Email Service

An email server is deployed in the server room to provide internal communication across all university zones. The email service uses SMTP for sending and POP3 for receiving messages.

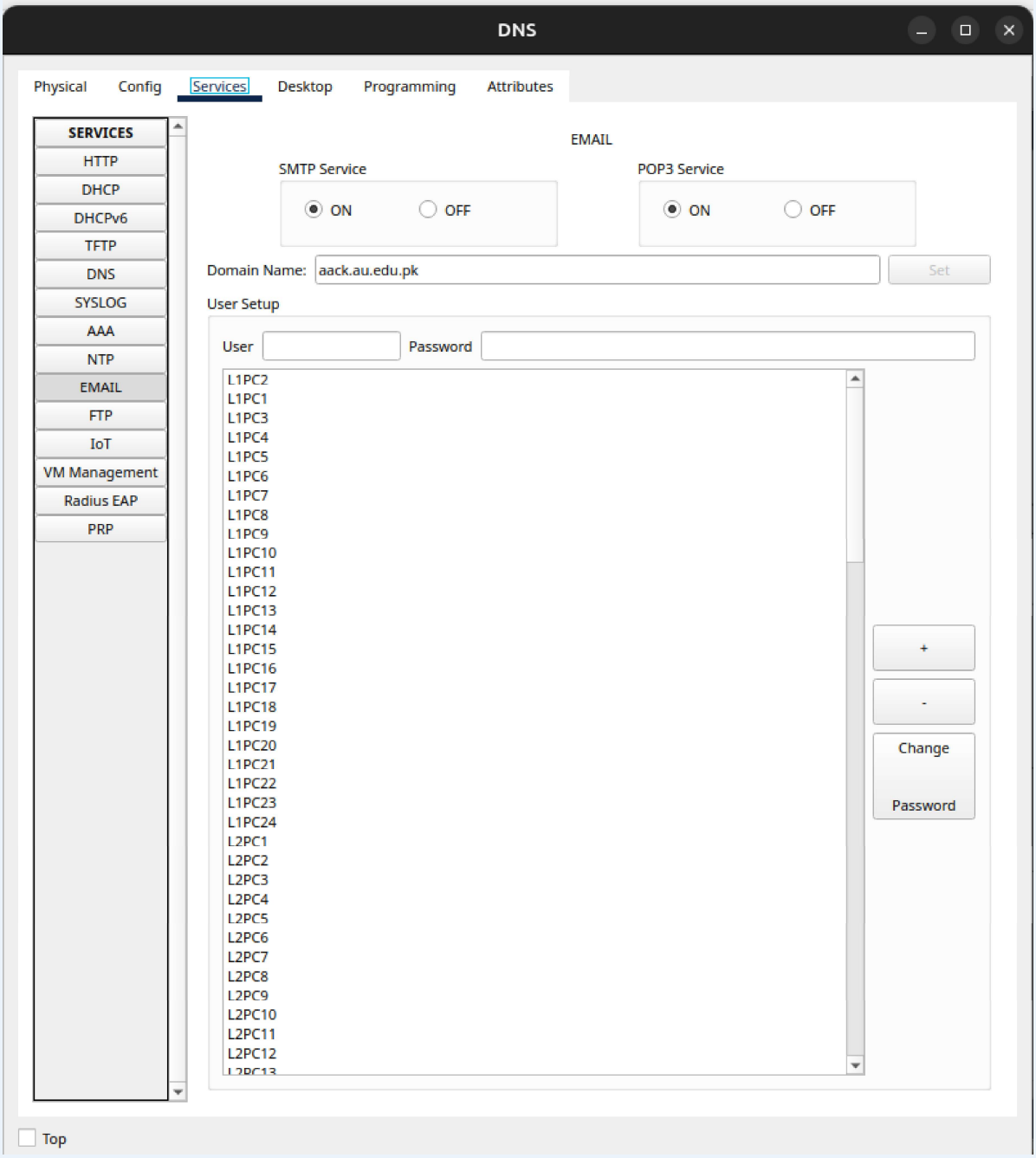
7.1 Email Server Settings

- Domain: aack.au.edu.pk
- SMTP Port: 25
- POP3 Port: 110
- Server IP: 192.168.100.10

7.2 PC Email Client Setup

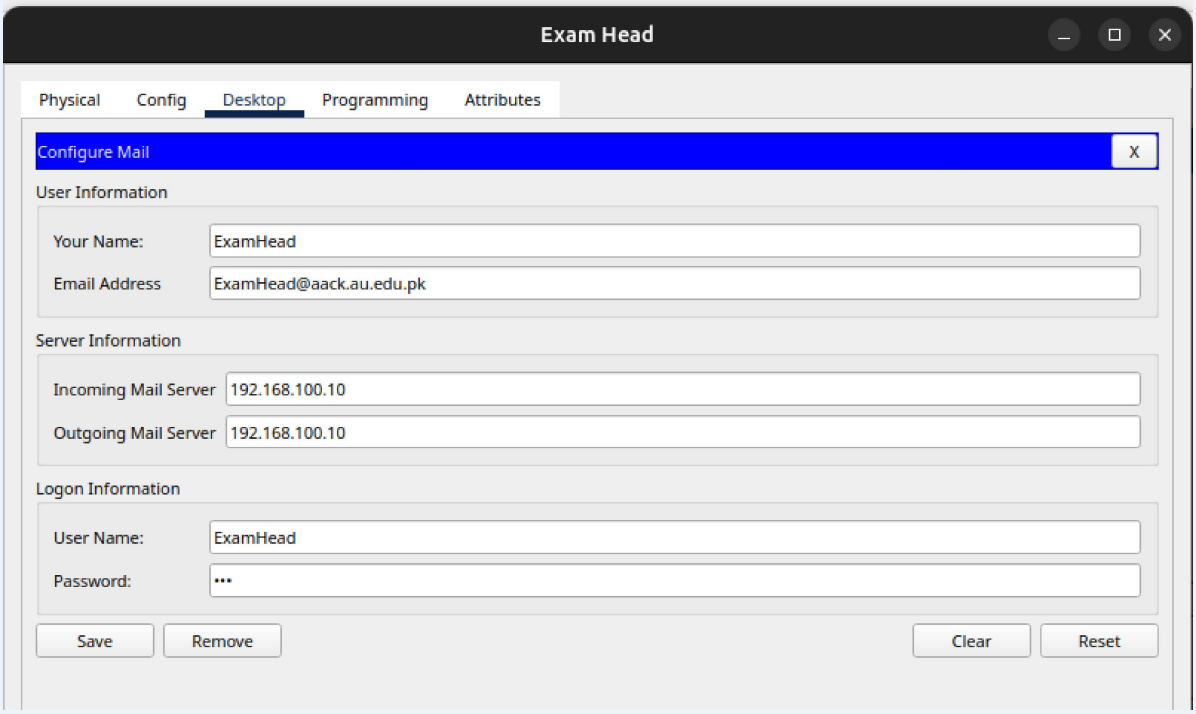
Each PC is configured with the following email client settings:

- Incoming Mail Server (POP3): 192.168.100.10
- Outgoing Mail Server (SMTP): 192.168.100.10
- Username: i.e.: L1PC1@aack.au.edu.pk || HODCS@aack.au.edu.pk



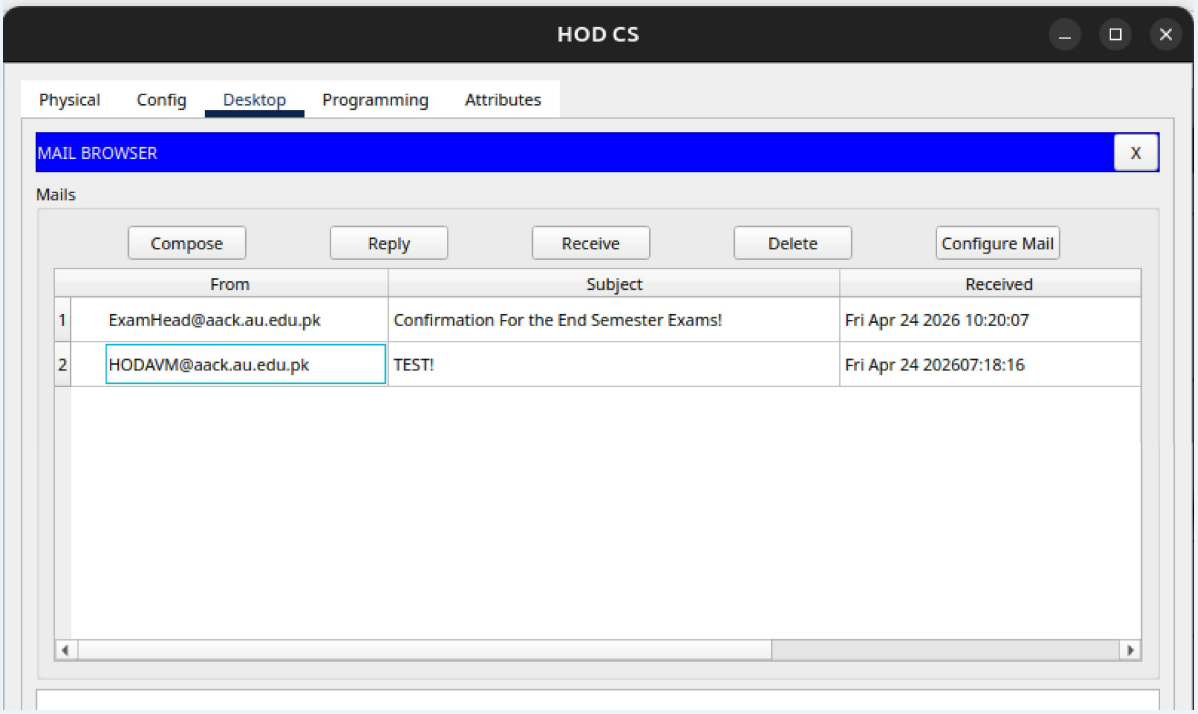
DNS User Setup and the POP3 & SMTP Service

Configuration Settings

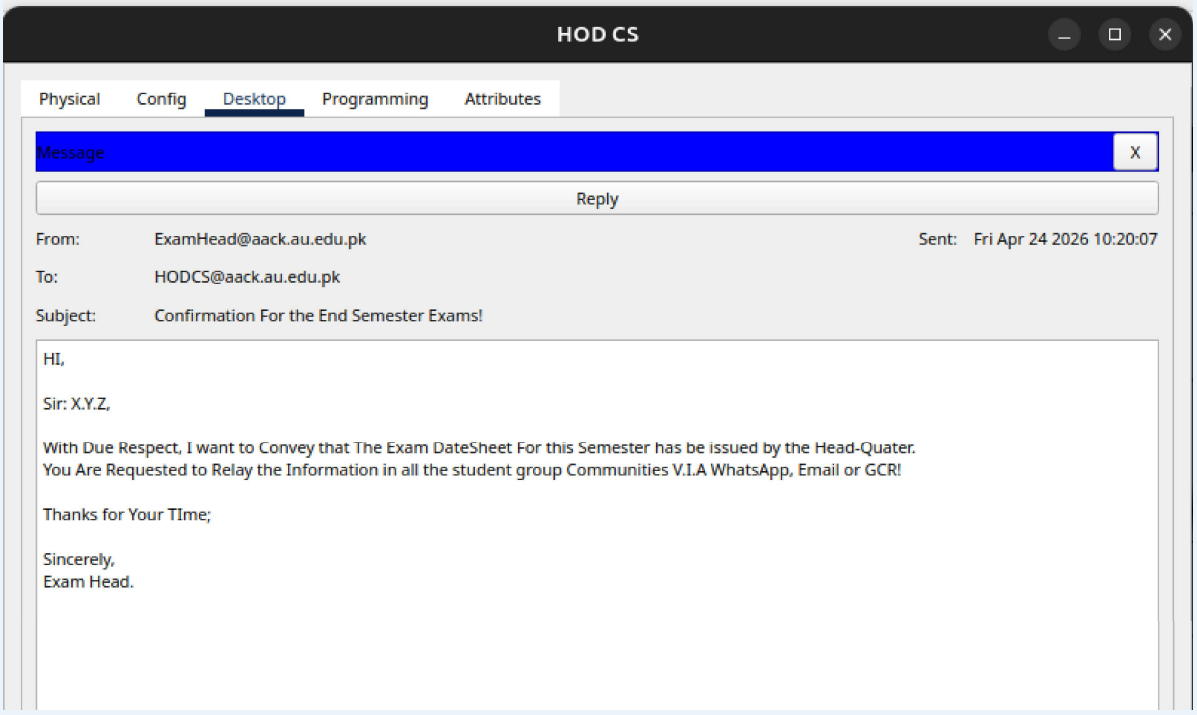


Configurations Done to ones PC

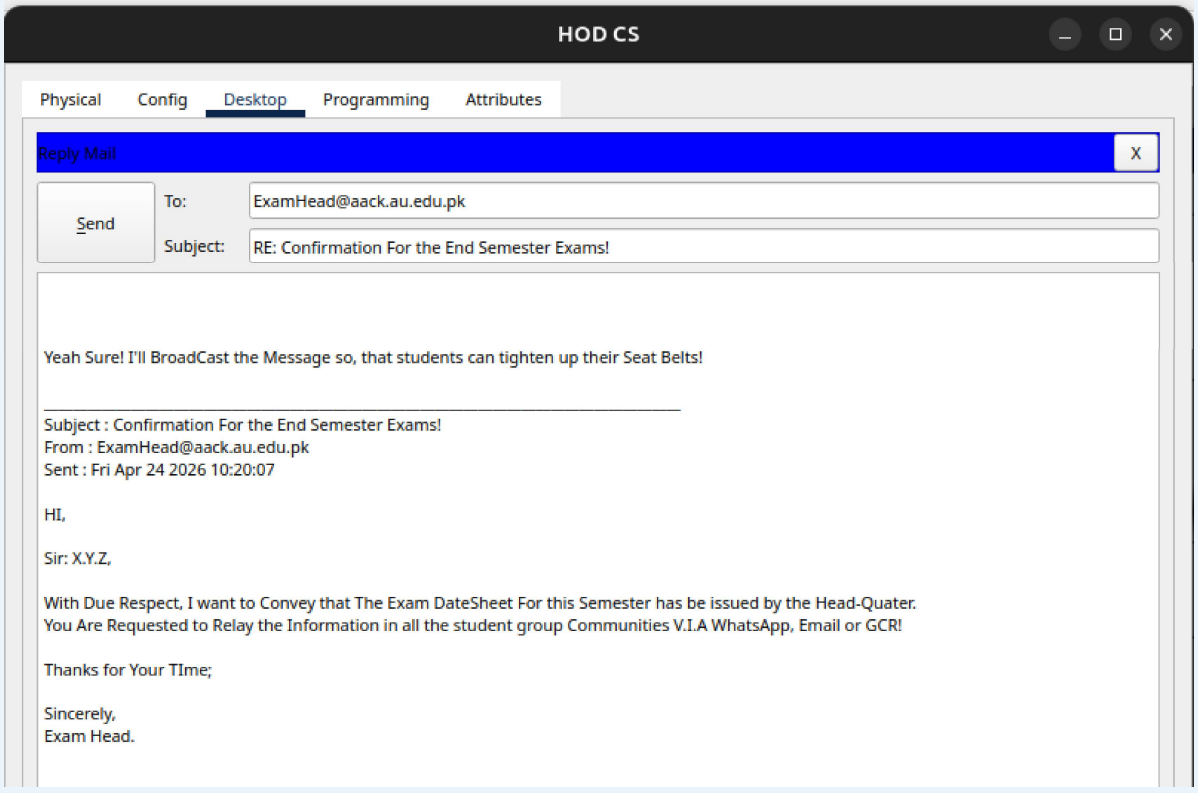
Email Communication Between Two PC's



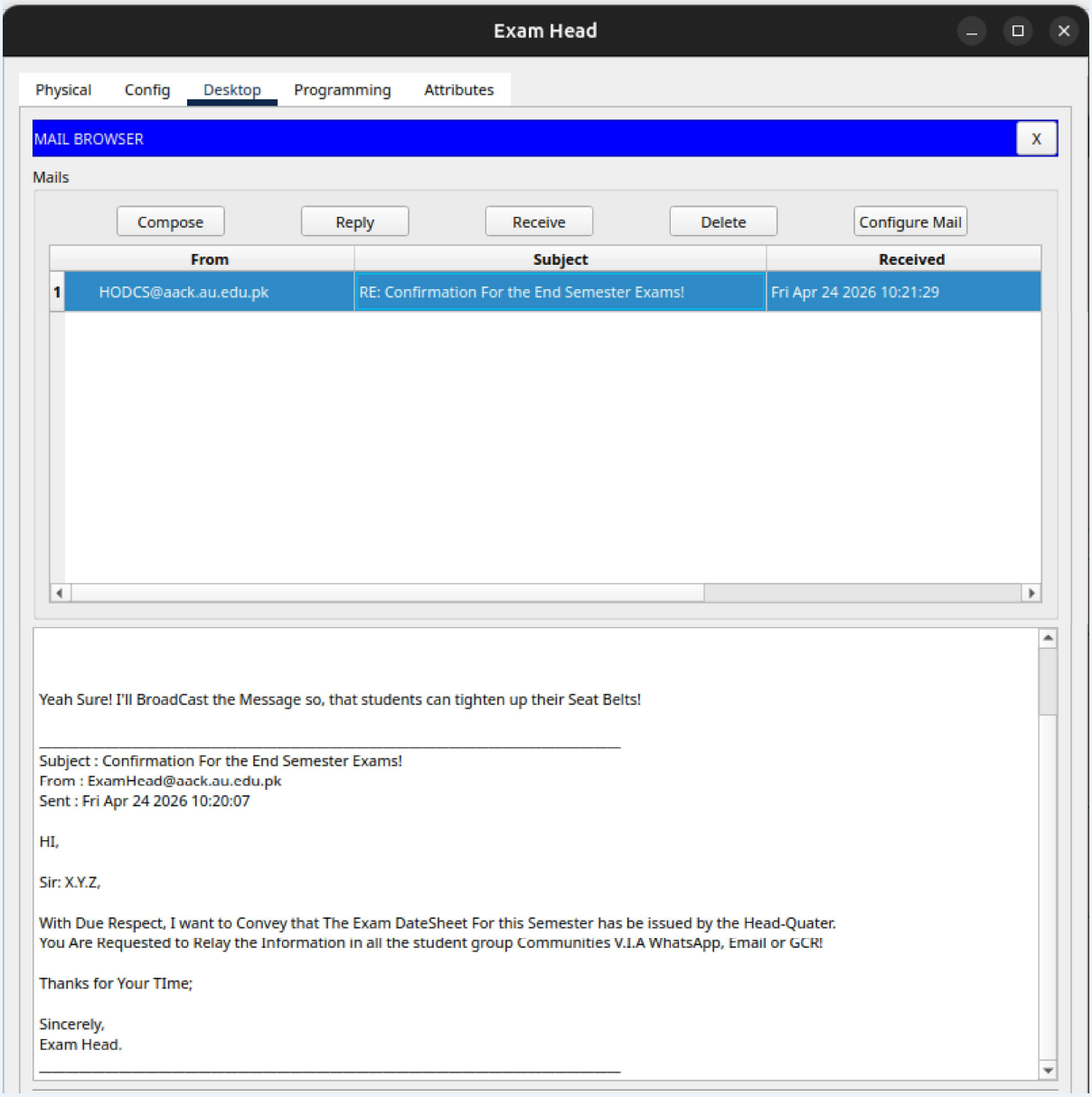
Email Recieved



Email Opened!



Replying Back to Email



Recieved Back Email

8. Access Control Lists (ACLs)

Access Control Lists are configured to enforce a realistic university access hierarchy. The policy ensures that sensitive offices like HOD and Exam Office are protected from unauthorized access while still allowing necessary communication.

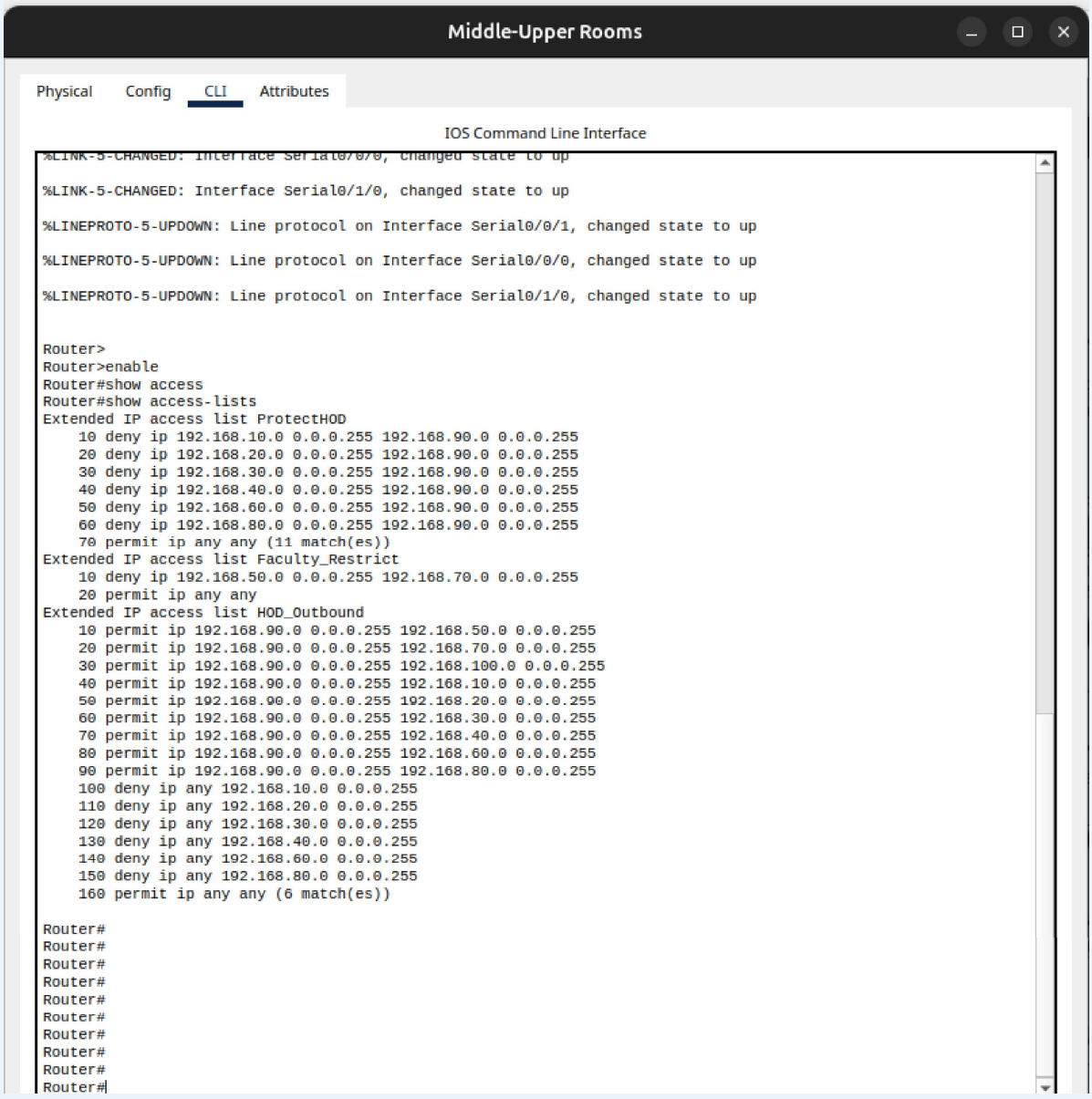
8.1 Access Policy Matrix

Source \ Destination	Server Room	Faculty Office	HOD Office	Exam Office
Labs / Classrooms	✅ Allowed	✅ Allowed	❌ Blocked	❌ Blocked
Faculty Office	✅ Allowed	—	✅ Allowed	❌ Blocked
HOD Office	✅ Allowed	✅ Allowed	—	✅ Allowed
Exam Office	❌ Blocked	❌ Blocked	✅ Allowed	—
Server Room	—	✅ Allowed	✅ Allowed	✅ Allowed

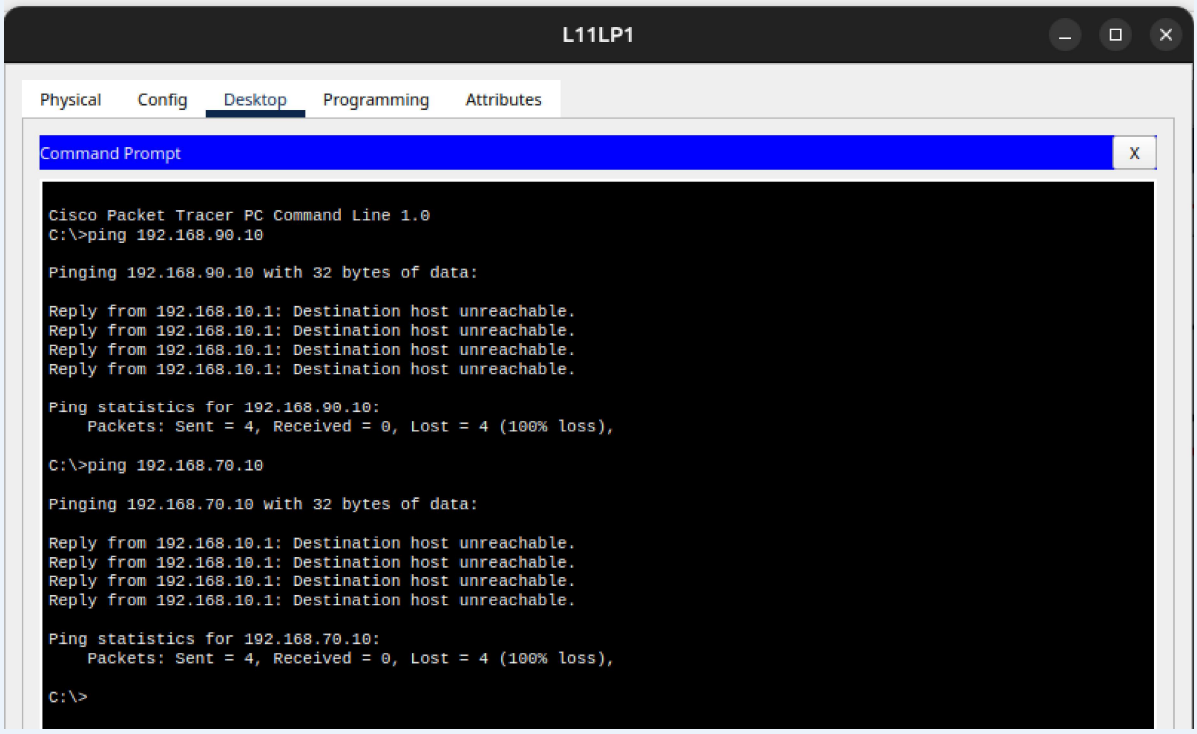
📌 **NOTE:** HOD Office can initiate contact with Labs and Classrooms (one-way). Labs cannot reply back to HOD — this models real university hierarchy where HOD broadcasts announcements but students cannot directly contact the HOD.

8.2 ACL Placement

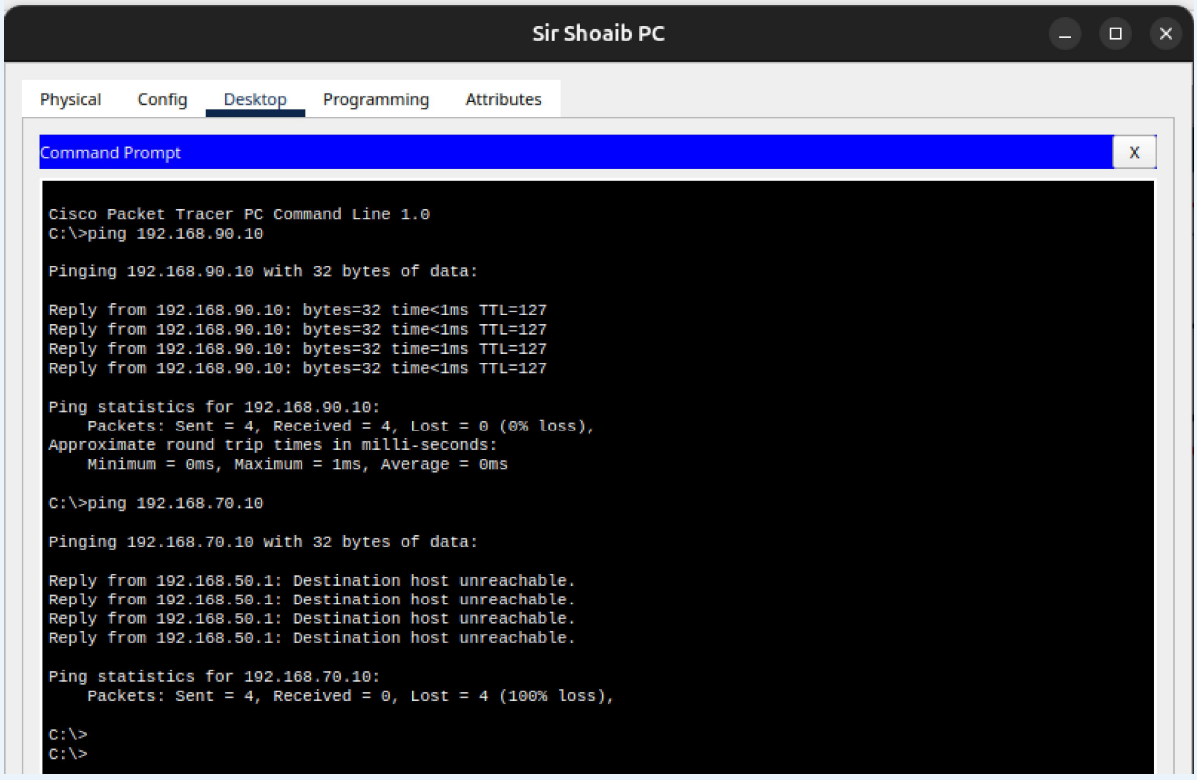
- ProtectHOD ACL → Applied inbound on Mid-Upper Router (GigabitEthernet0/1)
- ProtectExam ACL → Applied inbound on Mid-Lower Router (GigabitEthernet0/1)
- Left_Restrict ACL → Applied inbound on Left Router (Gig0/0 and Gig0/1)
- Right_Restrict ACL → Applied inbound on Right Router (Gig0/0 and Gig0/1)
- Top_Restrict ACL → Applied inbound on Upper Router
- Bottom_Restrict ACL → Applied inbound on Lower Router
- Faculty_Restrict ACL → Applied inbound on Mid-Upper Router (Gig0/0)
- HOD_Outbound ACL → Applied outbound on Mid-Upper Router (Gig0/1)



This Shows the IP access-lists groups on Middle Upper Router that are working to complete the meaning of ACLs



Lab PC Ping to HOD office and Exam Office but unable to reach



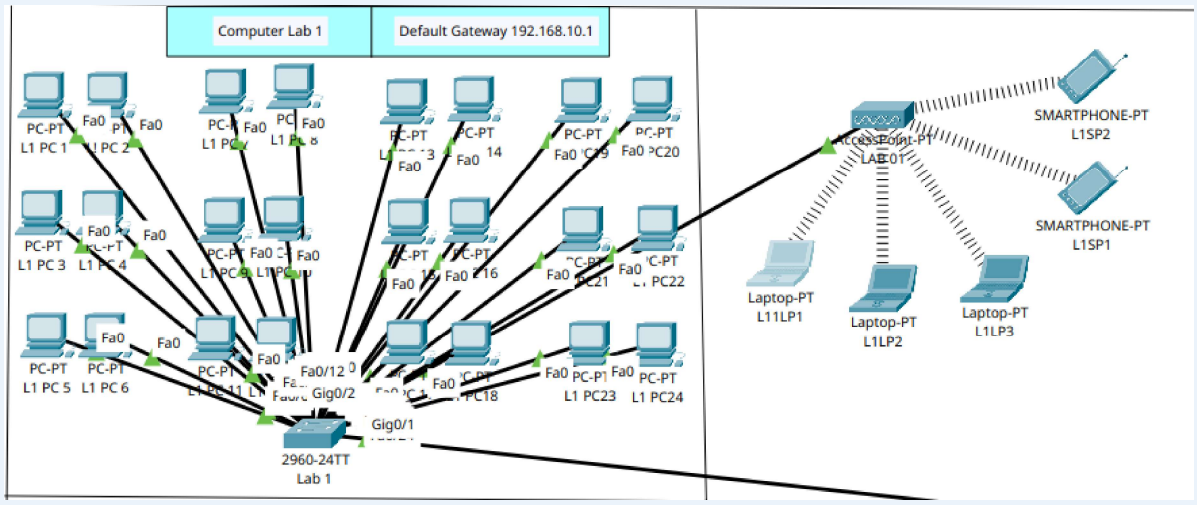
Faculty Member Pings to HOD (Success) & Exam (Fails)

9. Wireless Access Points

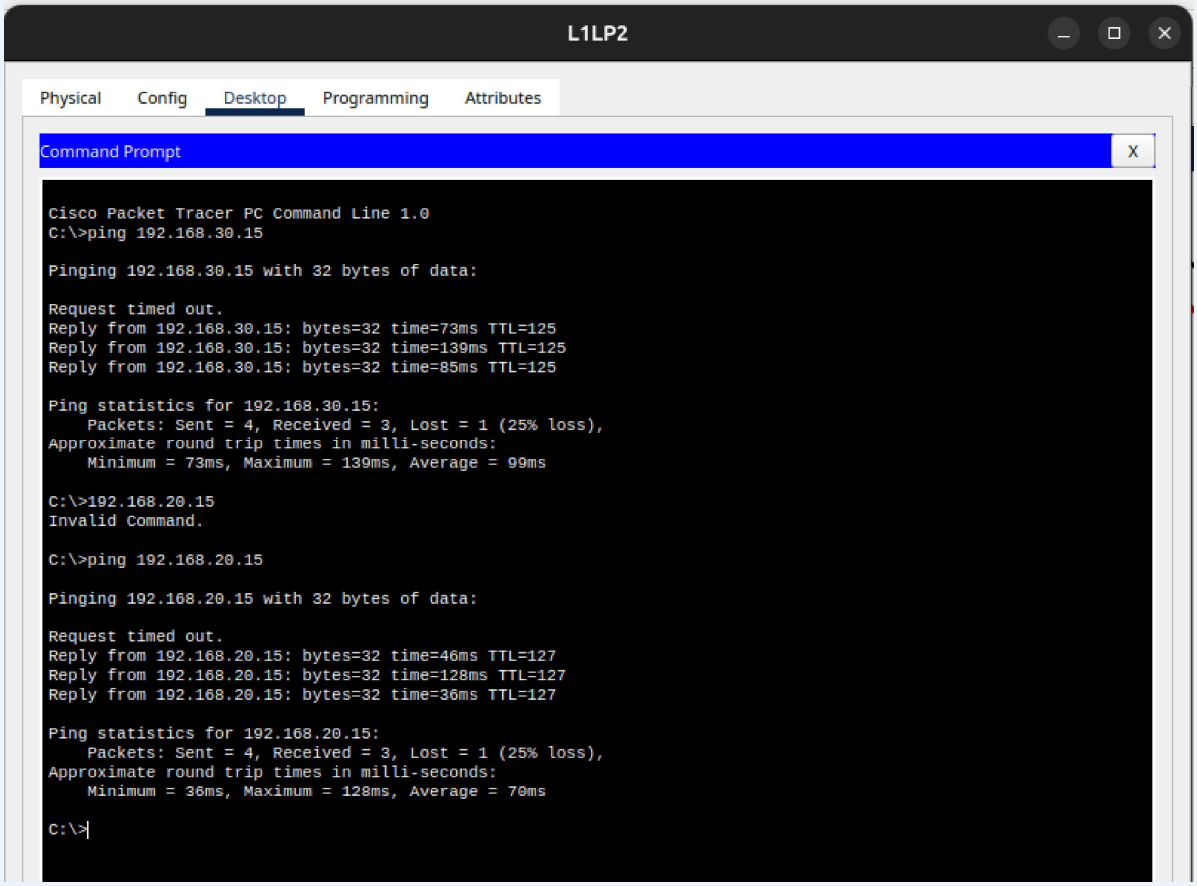
Wireless access points are deployed across the university building to provide Wi-Fi connectivity for laptops and mobile devices. Each access point is connected to the existing zone switch and shares the same subnet and gateway as wired devices in that zone.

9.1 Wireless Deployment

Location	Connected To	Subnet
Computer Lab Area	Left: Lab01 Switch	192.168.10.0
Exam Department	Middle-Lower: Exam Office	192.168.70.0



NOTE: We have only Added Two in whole network, because our main purpose was to demonstrate the Wireless Connectivity. So it didn't mattered to add to each and every Sub Sections.



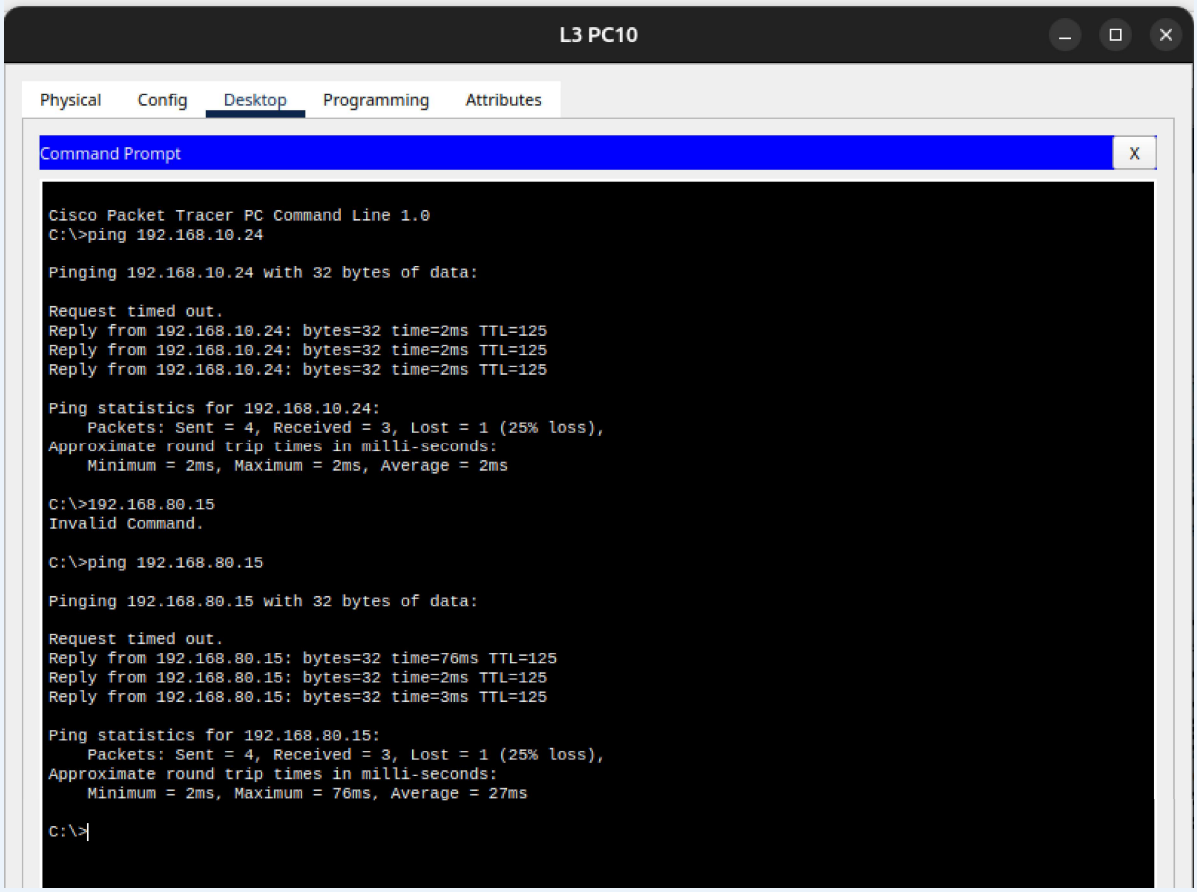
Lab01 Laptop 2: ping to Lab03 PC6 and Lab02 PC6

10. Testing & Verification

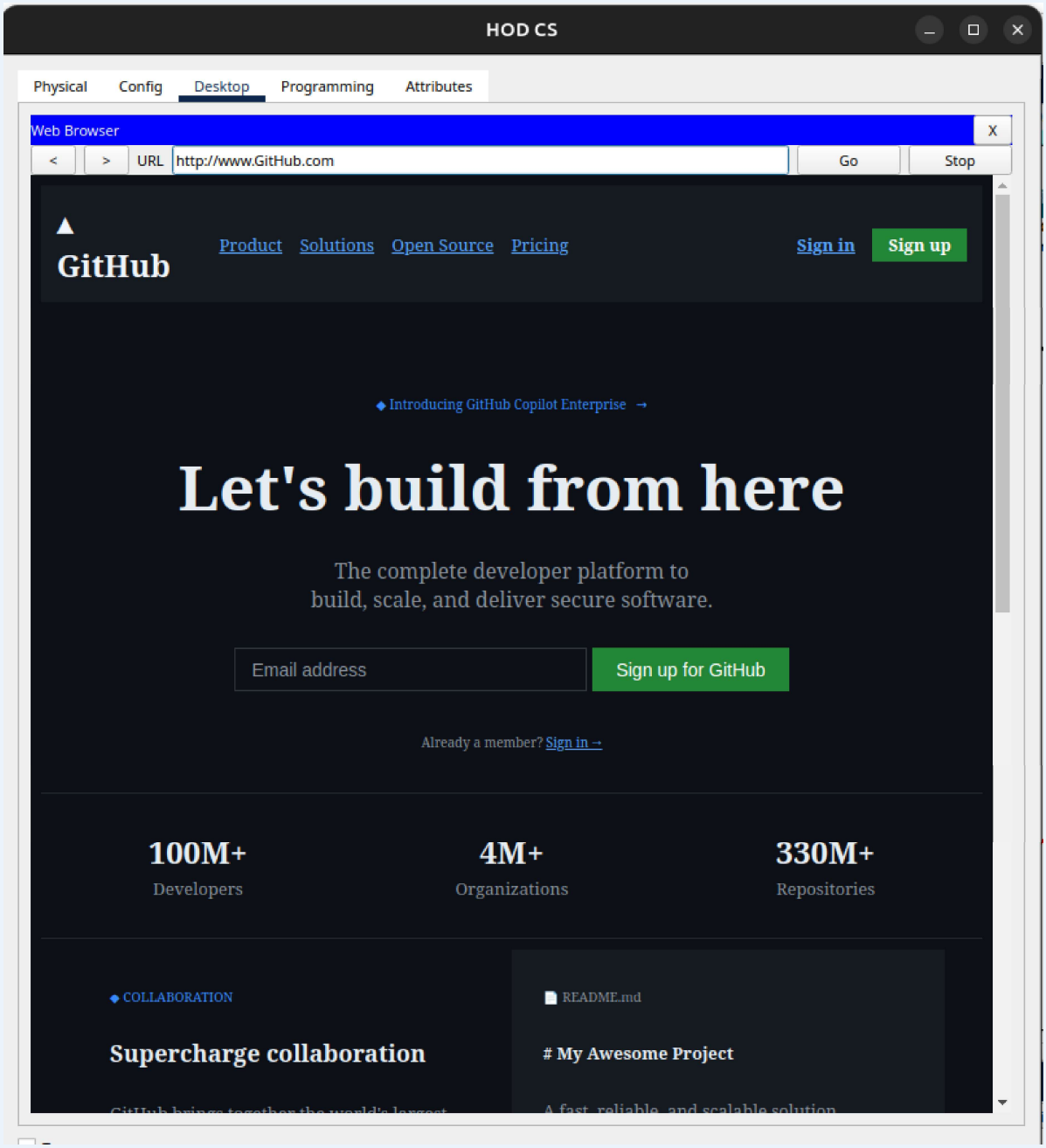
All network functionality was tested systematically after each configuration stage. Testing was done using Cisco Packet Tracer's simulation mode and CLI ping commands.

10.1 Connectivity Tests

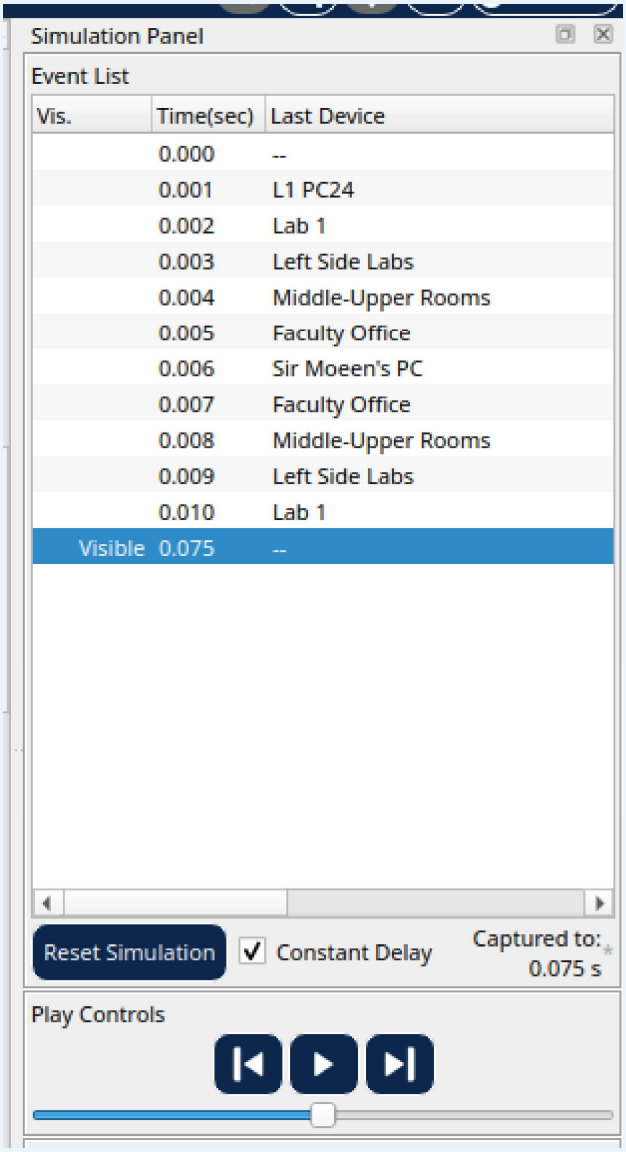
Test	Source	Destination	Result
Cross-zone ping	Lab 1 PC24	Faculty Office PC7	✅ PASS
Server access	Lab 1 PC24	IT Manager (192.168.100.10)	✅ PASS
DNS resolution	Any Lab PC	www.yahoo.com	✅ PASS
HTTP access	Any Lab PC	google.com web page	✅ PASS
ACL — Labs to HOD	Lab PC	HOD Office PC	✅ BLOCKED
ACL — Faculty to HOD	Faculty PC	HOD Office PC	✅ PASS
ACL — Exam to HOD	Exam Office PC	HOD Office PC	✅ PASS
Failover — Central DOWN	Lab 1 PC	Faculty Office PC	✅ PASS via Ring



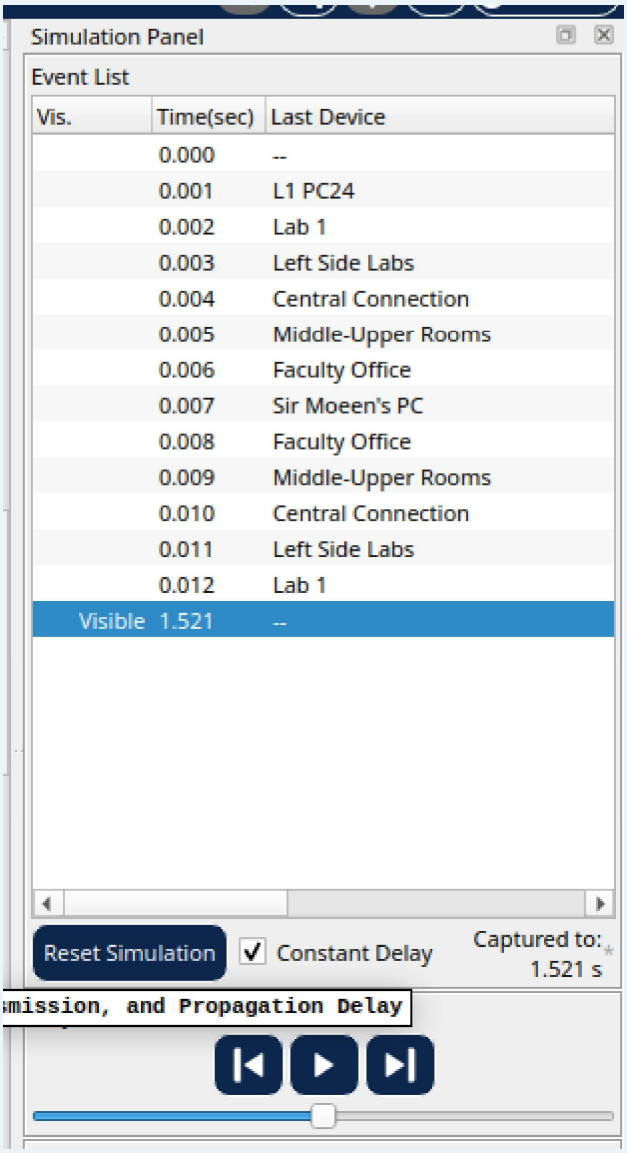
LAB3-PC10 (IP: 192.168.30.19) pings to Lower-section Classroom and LAB1-PC15



HODCS fetch the HTTP domain of www.GitHub.com



Upper One is the Message Pathway that it takes if the central connection router is down (BackUp Path)



The Main Path Message takes if the Central Connection router is UP. (Direct Path)

11. Challenges & Solutions

Challenge Faced	Solution Applied
HWIC-1GE-SFP module not linking up	Switched to HWIC-2T serial ports for router-to-router links
IR8340 ports were Layer 2 switch ports	Replaced with standard 2911 router as central
Broadcast ping 192.168.X.255 failing	Used directed ping to specific host IPs instead
Serial link IP overlap (wrong /8 mask)	Corrected subnet mask to 255.255.255.252 (/30)
Ring backup routes causing routing loop	Rebuilt all backup routes with correct ring order and next-hops
ACL blocking HOD reply traffic from Labs	Designed as intentional one-way policy — HOD initiates, Labs cannot reply
Teammate IP addressing errors	Manually corrected all IPs and gateways zone by zone

12. Conclusion

This project successfully demonstrates the design and implementation of a fully functional university campus network. The network achieves all primary objectives including inter-zone communication, fault-tolerant routing, DNS/HTTP services, email communication, access control, and wireless connectivity.

The hybrid Star-Ring topology with floating static routes ensures that the network remains operational even when the central router fails, with traffic automatically rerouting through the ring. The ACL policy models a realistic university access hierarchy, preventing unauthorized access to sensitive offices while maintaining full connectivity for legitimate communication.

Key technical concepts applied in this project include:

- Subnetting and static IP addressing across 10 distinct network zones
- Hierarchical router design with central backbone and zone distribution
- Serial DCE WAN links with /30 point-to-point subnets
- Floating static routes with Administrative Distance for fault tolerance
- Extended Named ACLs for granular traffic control
- DNS A-records and HTTP hosting for web services
- SMTP/POP3 email server for internal communication
- Wireless access point deployment for mobile connectivity

📌 **NOTE:** Future Scope: This network can be extended with OSPF dynamic routing, VLAN segmentation for enhanced security, QoS for traffic prioritization, and redundant server room connections for higher availability.

13. Appendix — Complete Subnet Reference

Router Interface Summary

Router	Interface	IP Address	Connected To
Central	Serial0/0/0	10.1.1.1	Left Side
Central	Serial0/0/1	10.1.2.1	Mid-Upper
Central	Serial0/1/0	10.1.3.1	Upper Side
Central	Serial0/1/1	10.1.4.1	Right Side
Central	Serial0/2/0	10.1.5.1	Mid-Lower
Central	Serial0/2/1	10.1.6.1	Lower Side
Left Side	Gig0/0	192.168.10.1	Lab 1 Switch
Left Side	Gig0/1	192.168.20.1	Lab 2 Switch
Left Side	Serial0/0/0	10.0.1.1	Ring → Mid-Upper
Left Side	Serial0/0/1	10.0.6.2	Ring → Bottom
Right Side	Gig0/0	192.168.30.1	Lab 3 Switch
Right Side	Gig0/1	192.168.40.1	Lab 4 Switch
Right Side	Serial0/0/0	10.0.3.2	Ring → Top
Right Side	Serial0/0/1	10.0.4.1	Ring → Mid-Lower
Mid-Upper	Gig0/0	192.168.50.1	Faculty Switch
Mid-Upper	Gig0/1	192.168.90.1	HOD Switch
Mid-Upper	Serial0/0/0	10.0.1.2	Ring → Left
Mid-Upper	Serial0/0/1	10.0.2.1	Ring → Top
Upper Side	Gig0/0	192.168.60.1	Classroom Switch
Upper Side	Serial0/0/0	10.0.2.2	Ring → Mid-Upper
Upper Side	Serial0/0/1	10.0.3.1	Ring → Right
Mid-Lower	Gig0/0	192.168.100.1	Server Room Switch
Mid-Lower	Gig0/1	192.168.70.1	Exam Office Switch
Mid-Lower	Serial0/0/0	10.0.4.2	Ring → Right
Mid-Lower	Serial0/0/1	10.0.5.1	Ring → Bottom
Lower Side	Gig0/0	192.168.80.1	Classroom Switch

Lower Side	Serial0/0/0	10.0.5.2	Ring → Mid-Lower
Lower Side	Serial0/0/1	10.0.6.1	Ring → Left

— End of Report —